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Catheter blood pump heart assist device

Abstract

A catheter blood pump includes a cannula with an expandable central portion having a proximal end, a distal end and a blood flow conduit therebetween. A flexible inlet portion at the distal end is provided with a plurality of spaced apart inlet struts configured to prevent an obstruction from entering, and a flexible outlet portion is provided with a plurality of spaced apart outlet struts operative to reduce a swirl velocity of blood, wherein the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts are fabricated from a collapsible shape memory material.

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6984392	12/2005	Bechert et al.	N/A	N/A
7010954	12/2005	Siess et al.	N/A	N/A
7011620	12/2005	Siess	N/A	N/A
7014417	12/2005	Salomon	N/A	N/A

7018182	12/2005	O'Mahony et al.	N/A	N/A
7022100	12/2005	Aboul-Hosn et al.	N/A	N/A
7027875	12/2005	Siess et al.	N/A	N/A
7037069	12/2005	Arnold et al.	N/A	N/A
7070555	12/2005	Siess	N/A	N/A
7122019	12/2005	Kesten et al.	N/A	N/A
7125376	12/2005	Viole et al.	N/A	N/A
7144365	12/2005	Bolling et al.	N/A	N/A
7150711	12/2005	Nusser et al.	N/A	N/A
7160243	12/2006	Medvedev	N/A	N/A
7172551	12/2006	Leasure	N/A	N/A
7175588	12/2006	Morello	N/A	N/A
7214038	12/2006	Saxer et al.	N/A	N/A
7229258	12/2006	Wood et al.	N/A	N/A
7238010	12/2006	Hershberger et al.	N/A	N/A
7241257	12/2006	Ainsworth et al.	N/A	N/A
7262531	12/2006	Li et al.	N/A	N/A
7264606	12/2006	Jarvik et al.	N/A	N/A
7267667	12/2006	Houde et al.	N/A	N/A
7284956	12/2006	Nose et al.	N/A	N/A
7288111	12/2006	Holloway et al.	N/A	N/A
7290929	12/2006	Smith et al.	N/A	N/A
7329236	12/2007	Kesten et al.	N/A	N/A
7331921	12/2007	Viole et al.	N/A	N/A
7335192	12/2007	Keren et al.	N/A	N/A
7341570	12/2007	Keren et al.	N/A	N/A
7381179	12/2007	Aboul-Hosn et al.	N/A	N/A
7393181	12/2007	McBride et al.	N/A	N/A
7393189	12/2007	Davis et al.	N/A	N/A
7469716	12/2007	Parrino et al.	N/A	N/A
7478999	12/2008	Limoges	N/A	N/A
7491163	12/2008	Viole et al.	N/A	N/A
7534258	12/2008	Gomez et al.	N/A	N/A
7547200	12/2008	O'Mahony et al.	N/A	N/A
7589441	12/2008	Kalsi et al.	N/A	N/A
7605298	12/2008	Bechert et al.	N/A	N/A
7619560	12/2008	Penna et al.	N/A	N/A
7632079	12/2008	Hershberger et al.	N/A	N/A
7633193	12/2008	Masoudipour et al.	N/A	N/A
7645225	12/2009	Medvedev et al.	N/A	N/A
7657324	12/2009	Westlund et al.	N/A	N/A
7682673	12/2009	Houston et al.	N/A	N/A
7722568	12/2009	Lenker et al.	N/A	N/A
7731675	12/2009	Aboul-Hosn et al.	N/A	N/A
7736296	12/2009	Siess et al.	N/A	N/A
7744566	12/2009	Pirovano et al.	N/A	N/A
7758521	12/2009	Morris et al.	N/A	N/A
7759521	12/2009	Bleuel	N/A	N/A
7766892	12/2009	Keren et al.	N/A	N/A
7780628	12/2009	Keren et al.	N/A	N/A

7785246	12/2009	Aboul-Hosn et al.	N/A	N/A
7811279	12/2009	John	N/A	N/A
7819833	12/2009	Ainsworth et al.	N/A	N/A
7820205	12/2009	Takakusagi et al.	N/A	N/A
7828710	12/2009	Shifflette	N/A	N/A
7841976	12/2009	McBride et al.	N/A	N/A
7842976	12/2009	Fujii et al.	N/A	N/A
7852996	12/2009	Lemke	N/A	N/A
7855316	12/2009	Meyer et al.	N/A	N/A
7878967	12/2010	Khanal	N/A	N/A
7918828	12/2010	Lundgaard et al.	N/A	N/A
7927068	12/2010	McBride et al.	N/A	N/A
7934912	12/2010	Voltenburg et al.	N/A	N/A
7935102	12/2010	Breznock et al.	N/A	N/A
7942804	12/2010	Khaw	N/A	N/A
7942844	12/2010	Moberg et al.	N/A	N/A
7955365	12/2010	Doty	N/A	N/A
7993259	12/2010	Kang et al.	N/A	N/A
7998054	12/2010	Bolling	N/A	N/A
7998190	12/2010	Gharib et al.	N/A	N/A
8012079	12/2010	Delgado, III	N/A	N/A
8025647	12/2010	Siess et al.	N/A	N/A
8052399	12/2010	Stemple et al.	N/A	N/A
8062008	12/2010	Voltenburg et al.	N/A	N/A
8079948	12/2010	Shifflette	N/A	N/A
8083503	12/2010	Voltenburg et al.	N/A	N/A
8110267	12/2011	Houston et al.	N/A	N/A
8114008	12/2011	Hidaka et al.	N/A	N/A
8123669	12/2011	Siess et al.	N/A	N/A
8142400	12/2011	Rotem et al.	N/A	N/A
8177703	12/2011	Smith et al.	N/A	N/A
8206350	12/2011	Mann et al.	N/A	N/A
8209015	12/2011	Glenn	N/A	N/A
8216122	12/2011	Kung	N/A	N/A
8235943	12/2011	Breznock et al.	N/A	N/A
8236040	12/2011	Mayberry et al.	N/A	N/A
8236044	12/2011	Robaina	N/A	N/A
8255050	12/2011	Mohl	N/A	N/A
8257054	12/2011	Voltenburg et al.	N/A	N/A
8257312	12/2011	Duffy	N/A	N/A
8262619	12/2011	Chebator et al.	N/A	N/A
8277470	12/2011	Demarais et al.	N/A	N/A
8317715	12/2011	Belleville et al.	N/A	N/A
8329913	12/2011	Murata et al.	N/A	N/A
8333687	12/2011	Farnan et al.	N/A	N/A
8348991	12/2012	Weber et al.	N/A	N/A
8364278	12/2012	Pianca et al.	N/A	N/A
8371832	12/2012	Rotem et al.	N/A	N/A
8376707	12/2012	McBride et al.	N/A	N/A
8382818	12/2012	Davis et al.	N/A	N/A

8388565	12/2012	Shifflette	N/A	N/A
8388582	12/2012	Eubanks et al.	N/A	N/A
8409128	12/2012	Ferrari	N/A	N/A
8414645	12/2012	Dwork et al.	N/A	N/A
8439859	12/2012	Pfeffer et al.	N/A	N/A
8449443	12/2012	Rodefeld et al.	N/A	N/A
8485961	12/2012	Campbell et al.	N/A	N/A
8489190	12/2012	Pfeffer et al.	N/A	N/A
8491285	12/2012	Haser et al.	N/A	N/A
8535211	12/2012	Campbell et al.	N/A	N/A
8540615	12/2012	Aboul-Hosn et al.	N/A	N/A
8545379	12/2012	Marseille et al.	N/A	N/A
8545380	12/2012	Farnan et al.	N/A	N/A
8579858	12/2012	Reitan et al.	N/A	N/A
8585572	12/2012	Mehmanesh	N/A	N/A
8591110	12/2012	Smith et al.	N/A	N/A
8591393	12/2012	Walters et al.	N/A	N/A
8597110	12/2012	Kammler et al.	N/A	N/A
8597170	12/2012	Walters et al.	N/A	N/A
8608635	12/2012	Yomtov et al.	N/A	N/A
8617239	12/2012	Reitan	N/A	N/A
8618239	12/2012	Gray	N/A	N/A
8684904	12/2013	Campbell et al.	N/A	N/A
8690749	12/2013	Nunez	N/A	N/A
8721516	12/2013	Scheckel	N/A	N/A
8721517	12/2013	Zeng et al.	N/A	N/A
8727959	12/2013	Reitan et al.	N/A	N/A
8734331	12/2013	Evans et al.	N/A	N/A
8734334	12/2013	Haramaty et al.	N/A	N/A
8784441	12/2013	Rosenbluth et al.	N/A	N/A
8790236	12/2013	LaRose et al.	N/A	N/A
8795576	12/2013	Tao et al.	N/A	N/A
8801590	12/2013	Mohl	N/A	N/A
8814776	12/2013	Hastie et al.	N/A	N/A
8814933	12/2013	Siess	N/A	N/A
8849398	12/2013	Evans	N/A	N/A
8944748	12/2014	Liebing	N/A	N/A
8992406	12/2014	Corbett	N/A	N/A
8998792	12/2014	Scheckel	N/A	N/A
9028216	12/2014	Schumacher et al.	N/A	N/A
9089634	12/2014	Schumacher et al.	N/A	N/A
9089670	12/2014	Scheckel	N/A	N/A
9162017	12/2014	Evans et al.	N/A	N/A
9217442	12/2014	Wiessler et al.	N/A	N/A
9308302	12/2015	Zeng	N/A	N/A
9314558	12/2015	Er	N/A	N/A
9327067	12/2015	Zeng et al.	N/A	N/A
9328741	12/2015	Liebing	N/A	N/A
9358329	12/2015	Fitzgerald et al.	N/A	N/A
9358330	12/2015	Schumacher	N/A	N/A

9381288	12/2015	Schenck	N/A	N/A
9421311	12/2015	Tanner	N/A	N/A
9446179	12/2015	Keenan	N/A	N/A
9675739	12/2016	Tanner et al.	N/A	N/A
9717833	12/2016	McBride	N/A	N/A
9872947	12/2017	Keenan et al.	N/A	N/A
10086121	12/2017	Fitzgerald et al.	N/A	N/A
10117980	12/2017	Keenan et al.	N/A	N/A
10195323	12/2018	Tiller et al.	N/A	N/A
10520025	12/2018	Peterson et al.	N/A	N/A
11033728	12/2020	Schenck et al.	N/A	N/A
11058865	12/2020	Fitzgerald et al.	N/A	N/A
11654276	12/2022	Fitzgerald et al.	N/A	N/A
11660441	12/2022	Fitzgerald et al.	N/A	N/A
11925795	12/2023	Muller et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/084,896 filed Oct. 30, 2020, which is a continuation of U.S. application Ser. No. 16/181,963, filed Nov. 6, 2018 and now issued U.S. Pat. No. 10,864,309, which is a continuation of U.S. application Ser. No. 15/633,189, filed Jun. 26, 2017, now U.S. Pat. No. 10,149,932, which is a continuation of U.S. application Ser. No. 15/176,620, filed Jun. 8, 2016, now U.S. Pat. No. 9,717,833, which is a continuation of U.S. application Ser. No. 13/618,071, filed Sep. 14, 2012, now U.S. Pat. No. 9,364,593, which is a continuation of U.S. application Ser. No. 12/945,594, filed on Nov. 12, 2010, now U.S. Pat. No. 9,364,592, which is a continuation of U.S. application Ser. No. 11/728,051, filed on Mar. 23, 2007, now U.S. Pat. No. 7,841,976, which claims the benefit of the filing dates of U.S. Provisional Application No. 60/785,299, filed on Mar. 23, 2006 and 60/785,531, filed on Mar. 23, 2006, the disclosures of which are hereby incorporated by reference herein in their entirety for all purposes. Further, the entire disclosures of U.S. Provisional Application No. 60/610,938, filed on Sep. 17, 2004, and U.S. patent application Ser. No. 11/227,277, filed on Sep. 15, 2005, are hereby incorporated by reference herein in their entirety for all purposes.

FIELD OF THE INVENTION

(1) The present invention relates to fluid pumping impellers, more particularly to expandable fluid pumping impellers. Still more particularly, the present invention relates to blood pumps such as left or right ventricular assist devices with an expandable impeller for treatment of heart disease.

BACKGROUND OF THE INVENTION

(2) Heart disease is a major problem in society, and claims many lives per year. After a heart attack, only a small number of patients can be treated successfully and non-invasively using medicines, such as pharmaceuticals. However, with sufficient mechanical assistance to the heart function, a majority of patients may recover from a heart attack, including even those with cardiogenic shock.

(3) In a conventional approach, a blood pump having a fixed cross-section is surgically inserted within the left ventricle of the heart and the aortic arch to assist the heart in its function. Surgical placement is required, since it is presently impractical or impossible to insert a pump of the size needed for sustaining adequate blood flow percutaneously. The object of the surgically inserted pump is to reduce the load on the heart muscle for a period of time, which may be as long as a week, allowing the affected heart muscle to recover while healing in a resting mode.

(4) Surgical insertion, however, can cause additional serious stresses in heart failure cases. Percutaneous insertion of a left ventricular assist device ("LVAD") therefore is desired. However, the conventional fixed cross-sectional diameter of such an LVAD cannot fit through the femoral artery of the leg in which it must travel to be positioned into the left ventricle. The maximum diameter of such a fixed diameter LVAD would have to be limited to approximately four millimeters for practical percutaneous insertion. This would limit the maximum pumped blood flow rate to approximately two liters per minute, approximately one-half the desired sustaining blood flow value for many cases. While the pumping rate can be increased by increasing the diameter of the device, particularly the diameter of the impeller, the size of the femoral artery is a limiting factor for percutaneous insertion. Hence, there is an urgent need for a pumping device that can be implanted through percutaneous insertion and yet provide the sustaining blood flow rates that conventional surgically implanted pumps provide.

SUMMARY OF THE INVENTION

(5) The present invention may be used as an LVAD, a right ventricular assist device ("RVAD") or in other situations that may benefit from a blood pump that is expandable in situ after being inserted into the body of a patient. The blood pump has an impeller design that allows compression and expansion of the impeller at the discretion of the operator. This compression/expansion feature allows for increased blood flow through the blood pump due to an increase, by expansion, of the impeller size, thereby producing a blood flow capable of sustaining human life without the need for significant contribution by the heart muscle. The blood flow provided is typically at least 4 liters per minute of blood, the flow that is usually sufficient to sustain life.

(6) The difference in using the blood pump as an LVAD as opposed to an RVAD is the location of the pump in the patient and the flow direction of blood through the pump.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is an end view of an embodiment of an impeller having three rows of blades (blade rows);

(2) FIG. 1B is a perspective view of the impeller of FIG. 1A;

(3) FIG. 2A is a highly schematic side elevational view of an embodiment of an impeller according to the present invention in its deployed configuration;

(4) FIG. 2B is a highly schematic side elevational view of the impeller of FIG. 2A in its stored configuration;

(5) FIG. 3 is a highly schematic side elevational view schematically illustrating the deployment of the impeller of FIG. 2A;

(6) FIG. 4A is an enlarged perspective view of a portion of a blade having an embodiment of a winglet;

(7) FIG. 4B is a cross-sectional view of the blade of FIG. 4A within a portion of a vessel;

(8) FIGS. 5A-5C are cross-sectional views of blades having exemplary winglet configurations;

(9) FIG. 5D shows cross-sectional views of exemplary winglet edge geometries

(10) FIGS. 6A-6D are end views of an impeller blade, further illustrating possible winglet configurations;

(11) FIG. 7A is an enlarged perspective view of a portion of an impeller according to the present invention having an indentation in the hub surrounding the proximate end of the blade;

(12) FIG. 7B is a cross-sectional view of the portion of the impeller shown in FIG. 7A;

(13) FIG. 8 is a stress-strain graph for a polyurethane material used to form an impeller blade;

(14) FIG. 9A is a perspective view of an embodiment of an impeller in a stored configuration within a storage housing;

(15) FIG. 9B is a perspective view of the impeller of FIG. 9A after emergence from the storage housing;

(16) FIG. 10 is a perspective view superimposing deployed and operational configurations of an embodiment of an impeller;

(17) FIG. 11 is a graph showing normalized average fluid shearing stresses as a function of normalized volume flow rates;

(18) FIG. 12 is a graph showing normalized average fluid shearing stresses as a function of tip gap sizes;

(19) FIG. 13 is a side view of another embodiment of an impeller according to the present invention;

(20) FIG. 14 is a perspective view of a blood pump according to the present invention;

(21) FIG. 15A is a side elevational view of the impeller portion of the blood pump of FIG. 14;

(22) FIG. 15B is a side elevational view of a cannula in which the impeller of the blood pump of FIG. 14 operates;

(23) FIG. 15C is a partial longitudinal cross-sectional view of the retainer sheath for use with the blood pump of FIG. 14;

(24) FIG. 16 shows highly schematic side views of different mesh designs; and

(25) FIG. 17 is a perspective view of the expandable portion of the cannula shown in FIG. 14 in the deployed state;

(26) FIG. 18 is an enlarged perspective view of the discharge or proximal end of the expanded cannula having a hexagonal mesh;

(27) FIG. 19 is an enlarged perspective view of the inlet or distal end of the expanded cannula showing a guide wire having a distal tip;

(28) FIG. 20 is an enlarged longitudinal cross-sectional view of an alternate embodiment of the distal end of the cannula forming a dilator;

(29) FIGS. 21A and 21B are side elevational views of the expandable portion of the cannula in stored and deployed configurations, respectively;

(30) FIGS. 22A and 22B are longitudinal highly schematic views of the blood pump of the present invention in the deployed and stored configurations, respectively, showing system components;

(31) FIG. 23A is a longitudinal cross-sectional view of the blood pump of FIG. 14 in its deployed configuration;

(32) FIG. 23B is a longitudinal cross-sectional view of the blood pump of FIG. 14 in its retracted position;

(33) FIG. 23C is a side elevational view of a cannula in which the impeller of the blood pump of FIG. 14 operates;

- (34) FIG. **24** is a longitudinal cross-sectional view of an alternate embodiment of a blood pump in its deployed configuration;
- (35) FIG. **25** is a side elevational view in partial cross-section showing a pre-parked sheath embodiment of the present invention; and
- (36) FIG. **26** is a highly schematic view showing the blood pump deployment in a patient.

DETAILED DESCRIPTION

(37) An impeller according to embodiments of the present invention includes a hub, and at least one blade supported by the hub. The impeller may have a deployed configuration in which the blade extends away from the hub, and a stored configuration in which the impeller is radially compressed, for example by folding the blade towards the hub.

(38) In some embodiments, the outer edge of a blade may have a winglet. The impeller also may have a trench or indentation proximate to a blade root to facilitate folding of the blade and/or to reduce shear stresses in the fluid flow induced by rotation of the impeller.

(39) Some embodiments of the present invention include impellers that do not radially compress, but retain a generally constant configuration. Impellers according to the present invention may be used in various applications, including improved blood pumps.

(40) Blade Rows

(41) Impellers according to embodiments of the present invention may include a plurality of blades which may be arranged in one or more blade rows positioned along the impeller hub. Figures IA and IB illustrate end and side views, respectively, of an impeller **100**. The impeller includes a hub **10**, and a plurality of blades **12** arranged in three blade rows. The first blade row includes blades **102** and **104**, the second blade row includes blades **106** and **108**, and the third blade row includes blades **110** and **112**.

(42) The provision of a plurality of blade rows facilitates the folding of the blades into a stored configuration as compared to the difficulty of folding a single helical blade extending a similar distance along the hub. Each blade row may include the same number of blades, for example, one to three blades. Alternatively, the number of blades in each blade row may differ. For embodiments in which there are more than two blade rows, the number of blades in at least one blade row may differ from the number of blades in other blade rows. The provision of a plurality of blade rows facilitates the achievement of larger values of fluid head or pressure rise than a single blade row, while allowing the impeller to be radially compressed while allowing into a stored configuration.

(43) One approach to impeller design provides an impeller having a long helical blade exhibiting a significant degree of wrap around the central hub. However, the three-dimensional shape of long helical blades limits the degree to which they can be folded without breaking or permanently deforming. By dividing a single helical blade into a plurality (two, three or possibly more) of individual blades, arranged in blade rows, the blades in each, row exhibit less wrap around the hub. Therefore, the individual blades may have an essentially two-dimensional shape which allows easier deformation during the storage process. The combination of two or more blade rows can produce the same flow and pressure as a single helical blade of similar axial extent. For example, individual blades may have a height-to-chord length ratio in the range of about 0.5-1.5, and a plurality of blade rows of such more easily folded blades may combine to provide a similar hydraulic efficiency as a longer serpentine blade. Further, in blood pumping applications, the use of a long serpentine blade may lead to separated flows, leading to thrombosis, which can be avoided using multiple blade rows.

(44) Hence, impellers according to some embodiments of the present invention may have multiple separate sets of blades, rather than a long, continuous helical blade. A continuous long helical blade is difficult to fold up against the hub, and by splitting a long blade into two or three shorter sections, the blade can be more easily folded into a cylindrical volume or space and subsequently deployed when desired.

(45) An impeller according to the present invention may include at least two blades arranged about

the circumference of the hub in a first blade row. The at least two blades may be positioned approximately $360/N^\circ$ apart from one another about the circumference of the hub, where N represents the total number of blades in the first blade row. The impeller alternatively may include a plurality of blades arranged in at least two blade rows, with each blade row including at least two blades. The at least two blades in the first row of blades may be positioned $360/N_{\text{sub}.1}^\circ$ apart from one another about the circumference of the hub and the at least two blades in the second row of blades may be positioned $360/N_{\text{sub}.2}^\circ$ apart from one another about the circumference of the hub, where $N_{\text{sub}.1}$ represents the total number of blades in the first row and $N_{\text{sub}.2}$ represents the total number of blades in the second row. $N_{\text{sub}.1}$ and $N_{\text{sub}.2}$ may be the same or different. The first and second rows of blades may be circumferentially offset relative to one another by $360/2N_{\text{sub}.1}^\circ$.

(46) Preferably, the number of blade rows is two or three. The blade rows may be interleaved (overlapping along the axial direction), which can increase performance but may increase the diameter of the impeller in the stored configuration. If blades are interleaved, they will tend to fold on each other in the stored configuration, increasing the stored diameter of the impeller.

(47) To minimize the stored diameter of the impeller, for example for blood pumping applications, the blade rows are preferably spaced apart along the hub, proximate to each other but not interleaved. For example, the spacing between blade rows may be less than the axial extent of each blade row along the hub axis. A larger blade row spacing allows shear wakes to decay between blade rows, but lengthens the impeller, making it more difficult to move the impeller along a curved path, such as along blood vessels to a desired location within a patient.

(48) Blade rows may also be clocked relative to each other, the clocking being an angular displacement between corresponding blades of each blade row, and in particular, an angular displacement between the trailing edge of a blade in one row and the leading edge of the corresponding blade in the next row. For example, an impeller may have at least a first blade row and a second blade row, the first and second blade rows including a similar number of blades. The blades of the first blade row may be angularly offset (clocked) relative to the corresponding blades of the second blade row. In blood pumping applications, the angular offset can be adjusted to reduce hemolysis. Blades may be clocked so that the leading edge of a following blade does not reside in the wake from a leading blade, and the clocking may be clockwise or counterclockwise to achieve this. Blade rows may be clocked relative to each other to avoid tandem blade effects, where the following blade resides wholly in the boundary layer or wake of the leading blade, so as to reduce shear stresses.

(49) Other Blade Parameters

(50) The amount of lean or tilt of the blades may be adjusted according to the blade stiffness. The blade lean may be a forward lean (toward the pressure face) of between about 30° and about 60° . A forward lean blade tends to deform so as to increase the angle of incidence of the fluid at the tip, and thus increase the load on the blades. Conventional propellers use a backward lean blade which tends to unload the blade tip under structural deflection. Hence, the use of a forward lean is unusual. Forward lean of flexible blades may be used to minimize the gap between a blade tip and the inside surface of a conduit in which the impeller operates. A backward lean may make it more difficult to control the size of the gap.

(51) The twist pitch angles of the blades also may be variable, for example for impeller operation in a conduit. The blade deviation angle may be in the range about 15 to 30 degrees to assist impeller operation at low Reynolds number operation (for example, less than 50,000 for the blade tip chord) within a conduit, hence reducing hemolysis in blood pump applications. The pitch at the tip, measured relative to a circumferential direction, can be appreciably less than the pitch at the root, to match slower fluid flow within a boundary layer near the inside surface of the conduit. Hence, the blade of an improved impeller for operation within a laminar flow profile within a cannula has a blade twist, the pitch being approximately matched to the flow profile, the blade tip having a

smaller pitch angle than the blade root. The blade may have a slightly humped appearance in a region of relatively rapid change in blade pitch.

(52) For external flow applications (not in a conduit), the twist in the blade pitch may be in the opposite direction as any boundary layer will tend to be closer to the hub.

(53) The root of a serpentine blade exhibits a geometric characteristic known as camber, the curvature of the airfoil section if laid out on a flat surface. By dividing a single long blade into two or more sections, the camber of the resulting partial sections can be limited, for example to values of less than 10%, for example 5%. In the latter case, the deviation of any section from a straight line will be on the order of five percent of the section chord length. For example, a serpentine blade having a camber of 15% may be divided into three sections, each of which will be substantially linear and more easily folded. The result would be an impeller having three blades, arranged in three blade rows, with performance similar to that of the serpentine blade. The wrap angle (from the leading edge to the trailing edge) can be limited to a maximum of about 30 degrees. Hence, blade rows may each contain substantially two-dimensional blades (compared with a single pumping efficiency), the blades serpentine blade of similar more readily folding against the hub than would a serpentine blade.

(54) The modulus of the blades may be lower for deforming the blades to the stored configuration of the impeller, which may correspond, for example, to strains of 100% 200%. In representative examples, the modulus for the larger blade deformations in the stored configuration may be about ten times less than the modulus for operational stresses. For example, the impeller may include blades formed from a material having a flexural modulus of about 10,000 psi for operational stresses and about 1,000 psi for storage deformations.

(55) FIG. 2A shows an impeller **200** in a deployed configuration, the impeller including a hub **210** and a plurality of blades **212**. Impeller **200** has a radius R_1 in the deployed configuration, as measured from the central longitudinal axis of hub **210** to the outermost blade tip. The deployed diameter is twice the deployed radius, and is the diameter of a circle described by the blade tip as impeller **200** rotates around the longitudinal axis of hub **210**. Also shown is a wall of a conduit **250** through which fluid flows relative to impeller **200**. Impeller **200** may be used as an axial pump, to pump fluid through conduit **250**. Alternatively, impeller **200** may be used as a motive force provider for a vehicle. For example, the impeller may power a boat, such as a jet-boat, or other water craft. In such example, conduit **250** could be a tube immersed in the water surrounding the vehicle, or there may be no conduit at all.

(56) FIG. 2B shows impeller **200** in a stored configuration, with blades **212** folded or otherwise deformed towards hub **210** and held in this stored configuration by a storage housing **260**. The radius $R_{sub.2}$ of storage housing **260**, which in this case defines the stored diameter of impeller **200**, is appreciably less than the radius $R_{sub.1}$ of the deployed impeller shown in FIG. 2A. A deformed impeller blade **212** may contact the inside surface of storage housing **260** at one or more locations.

(57) In embodiments of the present invention, the flexible blades **212** of impeller **200** can be folded or otherwise radially compressed such that the maximum diameter of the impeller in the stored configuration is approximately half, or less than half, the diameter of the impeller in the deployed configuration. Referring to FIGS. 2A and 2B, this corresponds to $R_{sub.2} \approx \leq (R_1/2)$. A ratio of $R_2 \approx \leq (R_1/2)$ is useful for blood pump applications, allowing a blood pump to deploy to a diameter of between about 6 millimeters and about 7 millimeters within a human body, while being non-surgically inserted with a diameter of between about 3 millimeters and about 4 millimeters. Other diameter ratios are useful for other applications.

(58) Impeller Deployment within a Conduit

(59) FIG. 3 is a schematic view illustrating the deployment of impeller **200**. Impeller **200** has hub **210** and blades **212**, and is retained in the stored configuration by storage housing **260**. Storage housing **260** may be a tube in which impeller **200** is stored prior to deployment. A drive shaft **230** is

used to rotate impeller **200**. The figure also shows an optional guide wire **280** within rotating drive shaft **230** which can be used to position impeller **200** at a desired location. The rotation of drive shaft **230** may also assist in deploying impeller **200**, for example through twisting the impeller out of storage housing **260** if the inner surface of the storage housing has a threaded texture.

(60) On the left of the figure, conduit **250** is shown into which impeller **200** is deployed for operation in its larger deployed configuration. Conduit **250** may represent any structure through which a fluid may flow relative to impeller **200**, such as a tube, catheter, cannula, or body vessel such as a blood vessel.

(61) Impeller **200** contained within its storage housing **260** may be deployed within a conduit, such as a utility pipe (water, gas, sewage, and the like), body vessel (such as a blood vessel), portion of a thrust unit for a vehicle, or other structure through which a fluid may flow. The impeller can be conveyed to a desired location within the conduit in a stored configuration, and then deployed to the deployed configuration. Impeller **200** can be deployed by urging the impeller axially out of storage housing **260**, for example using drive shaft **230** attached to the impeller. The impeller then unfolds into the deployed configuration using the stored potential energy of blades **212** in the stored configuration.

(62) The stored configuration facilitates conveyance of impeller **200** to the desired location, enabling it to be passed through openings which are smaller than the diameter of the impeller in the deployed configuration. To remove impeller **200** from the conduit after use, the impeller may be radially compressed back into the stored configuration, for example by urging the impeller back into storage housing **260**, such as by re-folding flexible blades **212** against hub **210**. The stored impeller may then be removed from the use location through an access hole having a dimension less than the diameter of the impeller in the deployed configuration. Hence, impeller **200** can be inserted in the stored configuration through a relatively small entrance hole into a conduit **250** of larger diameter.

(63) Although storage housing **260** is described above as a tube from which impeller **200** may be deployed by axial movement out of the storage housing, that need not be the case. Rather, storage housing **260** may itself be expandable or have an expandable portion, as described below.

Expansion of storage housing **260** would allow impeller **200** to deploy, such that the impeller would not need to be pushed axially out of the storage housing to achieve the deployed configuration. Thus, with reference to FIG. 3, conduit **250** may represent storage housing **260** in an expanded condition.

(64) Winglets

(65) Impellers according to the present invention may include at least one blade having a winglet. In some embodiments, all blades within a blade row may include such a winglet; other blades in the impeller may or may not include a winglet. A winglet may improve hydrodynamic performance of the impeller in the operating state, and may also reduce shear stresses that exist within the fluid being pumped. As a result, when the fluid being pumped includes biological structures such as cells, the degradation of such structures by the pumping action may be reduced.

(66) An impeller blade typically has a pair of opposed faces: a pressure face inducing relative motion of the fluid through pressure as the blade rotates through the fluid; and a suction face inducing fluid motion by suction. Typically, the pressure and suction faces are not planar, but rather are curved in the same general direction to define an airfoil shape. The blade also has a leading edge cutting through the fluid as the blade rotates, a trailing edge, and an outer edge (which may also be referred to as a blade tip or distal end of the blade). A winglet may extend in the direction of motion of the impeller (from the pressure face of the blade), in the direction opposite the direction of motion (from the suction face of the blade), or in both directions.

(67) FIGS. 4A and 4B show perspective and cross-sectional views, respectively, of a blade **212** of impeller **200** having a winglet **222** at its distal end. FIG. 4A shows the cross-section of the blade **212** where it joins winglet **222** as a dashed line, the winglet significantly enlarging the cross-section

of the blade at its distal end. FIG. 4B shows blade **212** and winglet **222** in cross-section, in which the winglet and blade form an approximate T-shape. As shown in FIG. 4B, the suction side of the blade is on the right, and the pressure side is on the left. If blade **212** has a thickness between the pressure face and the suction face at the distal end of the blade, winglet **222** may have a width between about 1 and 3 times the distal thickness of the blade, measured in a direction parallel to the blade rotation direction. If blade **212** has a chord length, winglet **222** may have a length approximately equal to the chord length.

(68) Winglets **222** are preferably aerodynamically smooth shapes having leading edges where flows impact the edges of the winglets, and trailing edges where flow is discharged from the winglet surfaces. Winglets **222** preferably have smooth aerodynamic cross-sections, generally in the direction of the mean flow, which is parallel to the flow direction along the blade tip surfaces. FIG. 5D shows possible leading edge geometries, including a radius edge **240**, a sharp edge **242**, and chisel edges **244** and **246**.

(69) Where impeller **200** rotates within a conduit **250** for fluid flow, the distal end of blades **212** in the deployed configuration, either with or without a winglet **222**, may be located proximate to the interior surface of the conduit, so as to define a tip gap **255** between the blade distal end and the inner surface of the conduit. The tip gap **255** may be about 10 to 50 percent of the maximum thickness of the distal end of the blade. In such circumstances, appropriately shaping the tips of the blades, such as by providing the tips with winglets **222**, can improve the quality of the flow field and reduce shear stresses. As shown in FIG. 4B, the winglet **222** may be proximate to the inner surface of conduit **250**, a configuration which may be used as a hydraulic bearing for an impeller **200**.

(70) For blood pump applications, simulations have shown that most hemolysis occurs at the blade tip, and that winglets lowered the hemolysis. Alternatively, the tip shape may be rounded to reduce hemolysis. A rounded blade tip reduces flow separation and turbulence at the tip compared with a squared-off tip, and winglets further reduce flow separation and hence turbulence in the wake of the blade tip as it moves relative to the fluid. By using a winglet **222**, the gap **255** between the tip of blade **212** and the inner surface of conduit **250** may be increased while retaining the performance of an impeller having a smaller tip gap but no winglets. This effect is analogous to retained lift near the end of winglet-equipped airplane wings. Fluid head losses are minimized for a tip gap in the range of about 0.10-0.15 times the maximum thickness of the distal end of the blade for blades without winglets, this range being expected to reduce hemolysis by reducing shear stresses in gap flows. Winglet-equipped blades show minimum fluid head losses at a tip gap of about 0.25-0.30 times the maximum thickness of the distal end of the blade. An increased tip gap reduces shear stresses for impeller operation in a conduit, and for blood pumping applications hemolysis is reduced compared with the use of a smaller tip gap.

(71) FIG. 5A shows a suction side winglet **262** extending from the outer edge of the suction face of blade **212**. This is a view from the leading edge, in cross-section, so that the blade rotates towards the direction of viewing. FIG. 5B shows a pressure side winglet **264** extending from the pressure face of blade **212**. The parameters may be similar to the suction side winglet. The function of the pressure side winglet is to reduce flow through the gap **255**. There is less effect of creating a hydrodynamic bearing, but the pressure side winglet “scrapes” low momentum fluid off the inner surface of the conduit **250** and prevents this fluid from entering gap **255** and subsequently being used in the core of a tip vortex. This can reduce shearing stresses in the bulk of the fluid flow.

(72) FIG. 5C illustrates a combined winglet **266** extending from the outer edge of both the pressure and suction faces of blade **212**. Embodiments of the present invention include the configurations shown in FIGS. 5A-5C. Numerical methods can be used to design the winglet configurations. Where the blade chord lengths are long and the blade has a significant helical extent, the geometry and shape of the blade tip and the winglet can become complex.

(73) FIGS. 6B-6D further illustrate winglet configurations, the blade supporting the winglet

retaining the same shape in these examples. FIG. 6A illustrates the outer edge shape **270** of a blade **212** not having a winglet.

(74) FIG. 6B shows a pressure side winglet extending from the outer edge of the pressure face of blade **212**, extending over portion **272**. The portion **270** of the winglet corresponds to the original outer edge shape of the blade shown in FIG. 6A.

(75) FIG. 6C shows a suction side winglet, the portion **274** extending from the outer edge of the suction face of the blade, and the portion **270** corresponding to the original outer edge shape of the blade. In some embodiments of the present invention, the pressure side of the blade will have a radius of approximately $\frac{1}{3}$ to $\frac{1}{2}$ the blade thickness or width. The extent of the winglet may be from $\frac{1}{2}$ to 3 times the blade thickness. A thickness approximately equal to the blade thickness is shown. The winglet is mostly positioned to the downstream half of the blade as shown. The purpose of this is to create a hydrodynamic bearing in which the outer face of the winglet is in close proximity to the inner surface of the conduit in which the blade is operating. With such configuration, the flow in the gap **255** between the winglet and the inner surface of conduit **250** is reduced in strength, and a tip vortex is less likely to form. This reduces shearing stresses in the fluid. Gap **255** can be between about 10 percent and about 25 percent of the base blade maximum thickness, and is an area that is mostly parallel to the fluid conduit **250**. It can be a cylindrical, conical or curved side cylinder where the radius is a function of the axial position of the blade element. Parameters for pressure side winglets and combined winglets (described below) may be similar.

(76) FIG. 6D shows a combined pressure side and suction side winglet extending from both the pressure face and the suction face of the blade, the portion **276** extending from the suction face, the portion **278** extending from the pressure face, and the portion **270** corresponding to the original outer edge shape of the blade.

(77) Features to Aid Stored Configuration

(78) Referring to FIGS. 7A and 7B, impeller **200** may have one or more structural features which aid achievement of the stored configuration. Such structural features may include one or more indentations proximate to the blade root to facilitate deformation of the blade into the stored configuration. An elongated indentation, such as a trench **282**, may be formed in the hub **210** of impeller **200** proximate at least part of the proximal end of the blade (the blade root, where the blade joins the hub). Trench **282** may be formed adjacent to one or both of the suction face and the pressure face of blade **212**. Preferably, trench **282** is formed in hub **210** parallel with and adjacent to the proximal end of the blade.

(79) The structural features may facilitate movement of the distal end of blade **212** towards hub **210**. For example, a trench **282** around some or all of the blade root can help reduce internal mechanical stresses in blade **212** when the blade is in the stored configuration, for example folded against hub **210**.

(80) In some embodiments, blade **212** may have a cross-section in the shape of an airfoil and the indentation may be a curved trench formed in the impeller hub **210** parallel to the proximal end of the blade. Other structural features which may aid achievement of the stored configuration include hinges (such as living hinges) with one or more indentations or cuts, not shown but known in the art, in the blade **212** and/or hub **210**; forming a portion of impeller **200** proximate the blade root from a more easily deformable material; and the like.

(81) The indentation may also be referred to as a “dillet,” and may include any undermining of the blade root. A dillet may be a trench proximate the blade root, for example having a depth between about 0.5 and about 1.5 times the blade width, and/or a width of a similar size range. A dillet can facilitate folding of the impeller blade towards the hub to achieve the stored configuration.

(82) The dillet may also reduce fluid shear stress and flow vortices in a fluid moving relative to impeller **200** as the impeller operates. In blood pumping applications, lower shear stresses lead to reduced hemolysis of the blood.

(83) Hub **210** may have dillets proximate both faces of blade **212**, one dillet facilitating folding of the blade (depending on the direction the blade is folded towards the hub), both dillets reducing the formation of a root junction vortex and hence reducing hemolysis in blood pumping applications. The dillet may be a horseshoe dillet, for example approximating the shape of a horseshoe vortex that would otherwise form at the blade root. Hence, dillets may be provided to reduce shear stresses, even for impeller blades that are not folded.

(84) Blade Materials and, Modulus

(85) Impeller **200** may be in the form of impeller body, including hub **210** and one or more unitary blades **212** formed from a single material. Alternatively, blades **212** and hub **210** may be formed from different materials. Preferably, blades **212** are flexible so that they can be deformed towards hub **210** in the stored configuration. Blades **212** may be formed in any way that allows expansion from a stored configuration to a deployed configuration, the deployed diameter of impeller **200** being larger than its stored diameter.

(86) Blades **212** may be formed from any material that permits the achievement of a stored configuration in which the blades are folded toward hub **210**. In that regard, the blades may be formed from a rubbery, elastic or other material having sufficient resilience to expand when the blades are no longer held in the stored configuration, such as when impeller **200** is deployed from a storage housing. The blades may be formed from polymer materials, such as polyurethane or other polymers having suitable elasticity properties. For medical devices such as blood pumps, biocompatible polymers are preferred. The average molecular weight may be chosen within a given range to obtain desired properties. Alternatively, the blades may be formed from other flexible polymers, from expandable foam optionally with a skin, or from other compressible or deformable materials including shape-change or shape-memory materials, and metals. Blades **212** may be formed with both a substantially rigid portion and a flexible portion, the blades being deformed towards hub **210** by deformation of the flexible portion. The flexible portion may include a hinge, such as a living hinge, a narrowed region, a material which is different from the material of the rigid portion, or other configuration.

(87) Blades **212** and (optionally) hub **210** may be constructed of a low modulus polymer, for example a low flexural modulus polyurethane (this term includes polyurethane ureas, which were used to form impellers according to the present invention). Impeller **200** may be a unitary structure, with the blades and hub formed as one from the same material, for example by molding a polymer.

(88) In some examples, blades **212** may have a stiffness approximating that of a thick rubber band. In such embodiments, the blades will have some stiffness, but will deform under operating loads. For example, the material forming impeller **200** may be chosen so as to have a linear modulus at operational stresses, allowing predictable deformation of the blades under load, and a non-linear modulus at the higher stresses used to deform the blades into the stored configuration.

(89) Impeller **200** may have blades **212** formed from a polymer, such as a polyurethane, having a flexural modulus (for operational stresses) between about 3,000 psi and about 30,000 psi, more preferably between about 5,000 psi and about 20,000 psi, and still more preferably between about 7,000 psi and about 10,000 psi. The modulus for operational stresses corresponds to deformations of the impeller during operation, which in some examples may correspond to strains of approximately 5%. The blade thickness may be reduced when using higher modulus materials to achieve the desired flexibility.

(90) Impeller blades **212** may, for example, occupy as much as 95% of the compressed volume of impeller **200** when the impeller is in the stored configuration.

(91) FIG. 8 is a stress-strain curve for a non-linear material that can be used to form impeller **200** according to the present invention. The left (low stress) filled circle corresponds to an impeller operating point (stress and strain under operating conditions) and the right (high stress) filled circle corresponds to the impeller stored configuration. The stress/strain relationship is approximately linear at the impeller operating point, so that deformations due to operational stresses can be

accurately predicted by numerical modeling. The stored configuration, in which blades **212** are folded against hub **210**, is within a high strain non-linear portion of the curve. This allows the stored configuration to be achieved without passing the material tensile failure point, and also reduces the stresses necessary to achieve the stored configuration. In example impellers **200**, the maximum material elongation in the stored configuration is about 75 percent.

(92) Preferably, a non-linear material, such as one having the characteristics of FIG. **8**, is used for blades **212**. This allows the blade material to be relatively stiff at operating loads, and relatively flexible at higher strains, such as when blades **212** are folded in the stored configuration. For example, the strain might be 1-10 percent at operating loads and 75 percent while folded, and the stress/strain curve may correspond to a higher modulus (e.g., 10,000 psi) at operating loads, and to a lower modulus (e.g., 1000 psi) at the higher loads associated with folding. The stress-strain curve may have two approximately linear regions with a sharp change in slope between the operating point strain and the folded strain.

(93) Impellers **200** may be fabricated from commercially available polyurethane polymers, for example having a modulus between about 5,000 psi and about 10,000 psi. Example impellers according to the present invention were fabricated as unitary bodies (including hub and blades) from elastomeric polymers. Example materials used include Conathane™ TU-901 (Cytec Industries, Inc., West Paterson, NJ), which had a modulus of about 10,000 psi for operational deformations; Conathane™ TU-701 (modulus of about 7,000 psi), and Hapflex™ 560 (Hapco Inc, Hanover, Mass), which had a modulus of about 5,000 psi. However, other polyurethanes, other polymers or other materials may be used.

(94) A polymer impeller **200** retained in the stored configuration for excessive time periods may not properly deploy, for example due to creep or electrostatic welding between adjacent polymer surfaces. Preferably, impeller **200** is retained in the stored configuration only as long as necessary to insert the impeller to a desired location. Hydrodynamic stress and forward lean may be helpful both to deployment and overcoming any hysteresis effect.

(95) Impeller **200** may deploy from the stored configuration due to stored potential energy associated with blade deformation towards hub **210** in the stored state. However, other stored potential energy may be used (for example, using shape memory materials). Depending on the application, external energy may be conveyed to impeller **200**, such as heat (for example, electrical heating of a wire or other structure), centrifugal forces, electromagnetic forces, gas jets, and the like to assist the deployment of the impeller.

(96) Impeller Fabrication

(97) Impeller **200** may be fabricated using molding, investment casting (for example, using a hard wax master), stereolithography, milling, or other techniques. Impellers **200** of the present invention have been fabricated using a flexible mold to avoid the presence of significant mold part lines.

(98) Very small impellers, approximately 6-7 mm in diameter in the deployed configuration, may be fabricated from a polymer (such as a polyurethane) and extracted from a precision mold. This allows production of impellers at very low cost. The flexible blades **212** allow the impeller to be extracted from a mold without becoming mold-locked, and allow the use of one-piece molds, instead of multi-part or split molds. This can be advantageous for producing impellers designed for pumping bio-fluids.

(99) Impeller Optimization

(100) Blade shapes can be optimized using standard computational fluid dynamics analysis (CFD). If the impeller material is not flexible, there is no deformation of the impeller when rotating. An improved method of optimizing an impeller **200** formed of a flexible material is to optimize the deployed configuration under operational stress (which may be termed the operational configuration). The impeller can be designed so that the operational configuration is optimized, which is not necessarily the same as the deployed configuration under no loading. A structural computation allows the determination of deformation under the load of operational stresses. Hence,

impeller **200** may have flexible blades **212** that deform into an optimized hydrodynamic shape when rotating and operating under design load conditions.

(101) The impeller blade **212** can be designed so as to minimize destruction of delicate particles (such as emulsion droplets, suspensions, and the like) within a biological structures fluid. A CFD model such as cells, may be used to simulate the through an intermediate be used to destruction shear stresses experienced by particles passing simulated impeller. Time integrations of shear stresses experienced by the particles may provide an estimated probability of cell in a biomedical application. A split blade design, in which there are a plurality of blade rows such as discussed above, reduces the residence time in which cells remain in intermediate shear stress regions, allowing an advantageous reduction in cell or other particle destruction compared with a single long helical blade.

(102) The impeller blade(s) **212** may deform during operation, and the optimum configuration of a blade may be achieved only upon deployment and rotation. For example, the optimal, design configuration of blade **212** may be achieved only with operational stresses. Hence, blade deformation in operation, due to flexibility of the blade, need not lead to reduced performance. Successful operation can occur even when impeller **200** exhibits significant deflections from a manufactured shape. The impeller can be manufactured with allowance for the deflection included in the design. The configuration of an impeller operating at a predetermined rotation rate, or within a predetermined operating range, can be optimized. Hence, in further embodiments of the present invention, the operational configuration of the impeller, including deformation due to operational stresses, is optimized.

(103) For blood pump applications, CFD optimization may be used to minimize flow velocity over blade surfaces (Reynolds number), vortex formation, flow jets, root junction flows, and to avoid formation of separated flows that may lead to thrombosis.

(104) Reynolds Number

(105) An impeller **200** according to the present invention can operate in a low Reynolds number conduit flow, where the conduit boundary layer comprises a majority of the flow in the conduit. The Reynolds number is the product of blade velocity and chord length, divided by the fluid viscosity. The Reynolds number varies with radius, and generally refers to the tip (distal end) of the blade unless otherwise stated. For example, the Reynolds number for operation of a conventional propeller may be on the order of millions, so that there is a turbulent flow transition as the fluid passes over the blade.

(106) Impellers **200** can be used with flows of small Reynolds number, less than 30,000 for the blade tip, for example the pumping of relatively viscous fluids at low velocity or flow rate. Impellers according to the present invention may operate with blade chord Reynolds numbers of between about 1,000 and about 30,000, preferably between about 2,000 and about 20,000, and more preferably between about 5,000 and about 20,000. The operation at such low Reynolds numbers corresponds to substantially laminar flow of the fluid over the blades. The reduced turbulence leads to reduced shear stress, and reduces hemolysis in blood pumping applications.

(107) Impellers **200** may also be used for flows of larger Reynolds numbers, such as from 100,000 to several million. Impeller diameters can be in the range of several millimeters (or less) to several meters, depending on the application.

(108) For operation in a conduit **250**, impeller **200** may be located within a fully developed laminar flow profile, for example at a distance of about 10-15 times the conduit diameter from the conduit inlet.

(109) A plurality of blade rows may be operated at a lower Reynolds number than a single longer serpentine blade having similar hydraulic efficiency. Each blade row may be separately optimized, for example to obtain substantially laminar flow. The blade rows may be clocked relative to one another to reduce hemolysis in blood pumping applications. Flow separations leading to thromboses may be avoided. Further, each blade row—may include a different number of blades,

for example 1, 2 or 3 blades. Hence, a plurality of blade rows may be used to reduce hemolysis in blood pumping applications compared to a single serpentine blade, while retaining similar or improved efficiency. For example, axial heads may be obtained similar to mixed flow pump heads. (110) For example, in an impeller **200** operated within a cannula as a blood pump, such as blood pump **600** described below, the blade chord Reynolds number of the first blade row was about 12,600, and was about 15,800 for the second blade row. This suggests that the flow was substantially laminar. In this case, the blades **212** may not exhibit a transition to turbulent flow over the blade surface (where the shear stress suddenly jumps to a higher value), which for blood pump applications leads to lower hemolysis.

(111) Hemolysis

(112) Hemolysis refers to the breakdown or destruction of red blood cells, releasing the hemoglobin contained therein. For blood pumping applications, hemolysis for a given impeller **200** can be estimated using equations known in the literature, and parameters discussed herein adjusted to reduce hemolysis.

(113) Both the hemolysis and platelet activation analyses can be conducted by combining the model of Garon and Farinas, “Fast Three-Dimensional Numerical Hemolysis Approximation,” *Artificial Organs*, 28(11):1016-1025 (2004), with empirical correlations of Giersiepen et al., “Estimation of Shear Stress-related Blood Damage in Heart Valve Prostheses—In vitro Comparison of 25 Aortic Valves,” *International Journal of Artificial Organs*, 13(5), 300-306 (1990). Giersiepen proposed empirical correlations for LDH-release by platelets:

(114) $\frac{LDH}{LDH}(\%) = 3.31 \times 10^{-6} t_{exp}^{0.77} \tau^{3.075}$, and for Rb-release by red blood cells:

(115) $\frac{Hb}{Hb}(\%) = 3.62 \times 10^{-5} t_{exp}^{0.785} \tau^{2.416}$, where t is in seconds and τ is in Pascals. The Garon and Farinas model, in short, provides a framework for any damage model of the form

$D = \gamma t \cdot \sup \cdot \exp \cdot \sup \cdot \beta \tau \cdot \sup \cdot \alpha$ by calculating the net flux of the parameter through a volume. So

$$(116) \quad D = (D_I) \quad \text{and} \quad D_I = \frac{I}{Q} \int_V dV$$

$\sigma = (\gamma) \cdot \sup \cdot (1/\beta) \tau \cdot \sup \cdot (\alpha/\beta)$.

(117) The variable 1 is the scalar form of the stress tensor referred to as the Von Mises Criterion and is specified as:

(118) $= [\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]]$ where σ_1 , σ_2 and σ_3 are the principal stresses. $g/100 \text{ L}$) is then: According to Garon and Farinas, the normalized index of hemolysis (in $NUH=100 \text{ Hb D}$, where Hb is the hemoglobin concentration in grams per liter, and the modified index of hemolysis (in parts per million) is:

$MIH=10MIH=10 \cdot \sup \cdot 6D$.

(119) By analogy, the platelet activation rate would be given by $PIA=PID$, where both PI and PIA are platelet concentrations, for example in thousands per microliter.

(120) Garon and Farinas only considered the hemolysis rate and only in laminar flow, and so the principal stresses are in reference to the laminar viscous stress tensor only. Garon and Farinas made no reference to how turbulent flow would be accommodated, but that discussion does take place in other research. Arvand et al. [“A Validated Computational Fluid Dynamics Model to Estimate Hemolysis in a Rotary Blood Pump,” *Artificial Organs*, 29(7):531-540 (2005)] actually advocated neglecting the Reynolds stress term in turbulent flow simulations in order to avoid “numerically caused variety” of the scalar form of the shear stress used in the hemolysis regression model, but the more conventional approach has been to use the effective shear stress (laminar plus Reynolds stress). See, for example, Gu et al., “Evaluation of Computational Models for Hemolysis Estimation,” *ASAIO Journal*, p. 202-207 (2005).

(121) Both approaches may be taken and compared for both and platelet activation analyses. The flow is simulated using computational fluid dynamics, the shear stress is determined from the

determined three-dimensional flow data, and the hemolysis may then be determined from the shear stress distribution.

(122) Stored Configuration and Deployment

(123) Impeller **200** may be stored in a storage housing, such as storage housing **260**, transported to a desired location in the stored configuration, and, once at the desired location, deployed into a deployed configuration. Rotation of the impeller then induces fluid flow at the location. For example, the impeller in the stored configuration may have a diameter approximately equal to or less than half the diameter of the impeller in the deployed configuration, the diameter of the stored configuration being generally defined by the inner diameter of the storage housing. The storage housing may be any assembly which acts to hold the impeller in the stored configuration, and may comprise a tube, sleeve, or similar structure inside which the impeller is stored prior to deployment.

(124) In the stored configuration of impeller **200**, blades **212** may be folded in towards hub **210**, or otherwise deformed or reconfigured so as to present a reduced diameter compared with the deployed configuration. Impeller **200** may be held in the stored configuration by storage housing **260**. In the stored configuration, the distal ends of blade **212** are closer to hub **210** than in the deployed configuration, and the stored diameter of the impeller can be significantly less than its deployed diameter. Stored diameters of about one-half of the deployed diameter or less are achievable.

(125) The storage housing need not have a fixed diameter, as does storage housing **260**, but may include a non-expandable portion, in which impeller **200** is stored, and an expandable portion, into which the impeller can be moved for deployment. Impeller **200** may then deploy within the expanded portion of the storage housing. The expandable portion of the storage housing may also have a stored configuration. For example, the diameter of the expandable portion in the stored configuration may be approximately half or less of its diameter in the expanded state. Alternatively, the entirety of the storage housing may be expandable such that impeller **200** does not have to be moved axially for deployment.

(126) Storage housings that are expandable or that include expandable portions may be held in a compressed state by a retainer sleeve, described below. Impeller **200** may be stored in a compressed configuration within the storage housing when the storage housing is retained in the compressed state. However, once the retainer sleeve is removed from that portion of the storage housing in which impeller **200** is located, the storage housing and impeller can expand to their expanded or deployed configurations. Impeller **200** may be deployed by urging the impeller blades **212** out of the confines of the storage housing, for example by pulling the retainer sleeve away from that portion of the storage housing overlying the impeller. In some embodiments, the retainer sleeve may be expanded in situ so as to allow impeller **200** to achieve the deployed configuration. Other methods of deploying impeller **200** will be clear to those skilled in the art.

(127) Impeller **200** may be deployed by various methods. For example, the storage housing may be expandable so as to have an expanded configuration when impeller **200** is in the deployed configuration, and a compressed configuration when the impeller is in the stored configuration. In such embodiments, the storage housing acts to radially compress impeller **200** in the stored configuration and allows the impeller to deploy when the storage housing expands. Alternatively, for storage housings that do not expand, impeller **200** may move axially out of the storage housing.

(128) FIG. 9A illustrates an impeller **300** in a stored configuration, showing blades **332** and **334** and hub **310**. Blades **332** and **334** are kept folded against hub **310** by the housing **360**. FIG. 9B shows impeller **300** pushed storage housing **360** and in the deployed configuration. Blades storage out of In the embodiment shown, impeller **300** has two rows of blades, as is seen more clearly in the deployed configuration, the first row including blades **332** and the second row including blades **334**.

(129) FIG. 10 shows an impeller **400** including a hub **410** and a plurality of blades shown in both the deployed and operating configurations. The figure allows comparison of the deployed

configuration under no load with the deployed configuration under operational stresses, when the impeller rotates at the operational rotation speed. In the deployed configuration under no load, the blades assume a first shape indicated by reference numbers **462A**, **464A** and **466A**. When rotating in a fluid, the blades deform to an operational configuration indicated by reference numbers **462B**, **464B** and **466B**. Impeller **400** may be designed so that the flexible blades deform into an optimized hydrodynamic shape when rotating and operating under design load conditions.

(130) In general, the blades deflect forward as the lift on the blades is such that they create thrust, a force directed towards the left side of the figure, moving the blades toward the left side of the picture. The leading edge of the second blade row is obscured. In this example, there are two blade rows, each with two identical blades. For example, the first blade row includes blade **462**, shown in an operating configuration at **462B** and under a no load condition at **462A**. The leading edge of each blade transitions smoothly into the trailing edge at the maximum blade radius. For a hub and blades formed from the same polymer, simulations showed that the hub also deflects slightly in a rotational manner, with the second blade row rotated at the root compared to the first blade row.

(131) FIGS. **11** and **12** illustrate optimization for fluid shear stress for an example impeller having a design similar to impeller **400** shown in FIG. **10**. The distal ends of the impeller blades move proximate to the interior surface of a cylindrical conduit such that the tip gap between the blade distal end and the inner surface of the conduit is about 10 to 50 percent of the maximum thickness of the distal end of the blade.

(132) The curves are double normalized, the design point values both being 1.0, the scales being read as percent of design flow and a factor times the value of stress at the design point. For example, FIG. **11** illustrates that at 70 percent of the design flow, the shear stress is 1.3 times the value at the design condition. FIG. **12** shows that making the tip gap smaller than the design value makes the shear stress higher, whereas making the gap bigger than the design value reduces the shear stress by a smaller factor. Therefore, the fluid shear stress can be reduced to lower hemolysis in blood pumping applications, without significantly compromising pumping efficiency.

(133) FIG. **13** illustrates an impeller **500** including a hub **510**, and two rows of blades having two blades each. The first row includes blades **582** and **584**, and the second row includes blades **586** and **588**. Impeller **500** has highly curved leading and trailing edge lines where the blade pitch angles are adjusted for local values of relative flow angle. Impeller **500** was designed for operation inside a conduit, such as a cannula having a laminar flow profile, the flow rates being lower near the inside surface of the conduit. This illustration shows the design elements of a low Reynolds number impeller for use in a left ventricular assist device (LVAD) in which the thickness of the boundary layer on the fluid conduit walls approaches the diameter of the conduit. The Reynolds number for a blood pump application was determined to be in the range of 10,000-20,000.

(134) Expandable Impeller and Expandable Cannula

(135) In some embodiments, an expandable impeller is used together with a cannula which may or may not have an expandable portion. If the impeller is not stored in an expandable portion, the impeller must be moved axially for expansion to its deployed configuration. If the impeller is stored in an expandable cannula or in an expandable portion of a cannula, the impeller expands into its deployed configuration with the expansion of the cannula. This combination may be used in improved blood pumps, such as an improved left ventricular assist device (LVAD).

(136) For example, a cannula may be provided that has expandable and non-expandable portions, and the impeller may be stored within, or proximate to, the non-expandable portion. The impeller can be urged out of the non-expandable portion of the cannula into an expanded portion of the cannula. The stored potential energy within the flexible blades of the impeller would then induce self-deployment of the impeller, and the cannula may also self-expand through stored potential energy. The expanded cannula then may have the role of a fluid conduit through which fluid flows when the impeller is rotated. An example of such system is blood pump **600** described below. An expandable cannula and impeller may both be stored within a retainer sheath and deployed together

when urged out of the retainer sheath, as is also described below.

(137) Applications

(138) Impellers according to the present invention may be used for a variety of applications, including an axial pump for a fluid (gas or liquid), a motive force for a vehicle, or other applications. Applications of the improved impellers according to embodiments of the present invention include pumps for chemical engineering, propellers for airborne or maritime vessels, water pumps, and the like.

(139) Impellers according to the present invention may be attached to one end of a flexible drive shaft. A torque applied to the other end of the drive shaft is then used to rotate the impeller. The torque may be applied by a rotating member, such as a motor.

(140) Blood Pump

(141) As noted above, impellers according to the present invention are well suited to blood pumping applications, including as a left ventricle assist device, as a right ventricle assist device, for pumping blood to other organs, and the like.

(142) For blood pumping applications, the impeller may operate within the laminar flow profile of a cannula flow, so that the blade pitch preferably varies with radius to match the flow profile. An impeller with two blade rows, such as impeller **500** illustrated in FIG. **13**, feature in the second row blades that may have a groove-like takes a helical path from the leading edge to the trailing edge. This arises due to variations in the span wise loading, and allows an axial flow pump using this impeller to achieve a head rise similar to that of a mixed flow pump.

(143) Computational fluid dynamics analysis shows that an axial blood pump—including an expandable impeller with two blade rows was suitable for use in a left ventricular assist device (LVAD). The impeller may be compressed and packaged into a storage housing, such as a tube, cannula, or other structure, for insertion into an object. For an object such as a living subject, the diameter of the storage housing can be about three to four millimeters or less. Having inserted the device, the impeller can be deployed in situ into a geometry that may be about six to seven millimeters in diameter. The impeller then can be rotated using a flexible drive shaft coupled to a drive motor external to the subject. Such impellers may be capable of pumping 4 Lpm (liters per minute), and more preferably 5 Lpm or greater, for example in a left ventricular assist device (LVAD).

(144) In a representative example of such a device, the impeller may rotate at about 30,000 RPM. The impeller may include two or more airfoil shaped blades that form an axial flow pump, and may be positioned using a guide wire. The guide wire may run within a hollow center of the flexible drive shaft, and the hollow center may also convey saline solution or other fluid for infusion, cooling and/or lubrication purposes. The guide wire may be removed, if desired. Implantation into a living subject may be achieved without surgical intervention through an insertion cannula having a diameter of about 3-4 mm. For example, a device including an impeller and a cannula may be inserted in a stored configuration through an insertion cannula in the femoral artery, the impeller and cannula then deploying (expanding radially) to approximately twice the stored configuration diameter when located at a desired location, such as proximate to the aortic valve.

(145) For medical implantation, a drive shaft comprising a metal braid, a polymer braid or a composite material braid may be used, and the drive shaft diameter may be on the order of 1.1 to 2 millimeters, and may be hollow to allow the guide wire to pass therethrough.

(146) An impeller according to the present invention can be operated within a cannula, and a flow of 5 Lpm at 100 mm Hg net pressure rise obtained (220 mm Hg across the impeller with pressure losses elsewhere). These parameters are well suited to blood pumping applications, for example within a left ventricular assist device (LVAD).

(147) A blood pump **600** for use in blood pumping applications, such as those noted above, is shown in FIG. **14**. Blood pump **600** may be broken down into three main parts as shown in FIGS. **15A**, **15B** and **15C**. It should be noted, however, that these features may be combined to produce

devices according to the present invention that are intended for applications other than blood pumping applications.

(148) The first part, shown in FIG. 15A, is an impeller **605** with a drive shaft **630** for implementing a rotational drive to the impeller. Impeller **605** includes a hub **610** and a plurality of blades **612**, and may have any or all of the features of the impellers described above. Hub **610** and drive shaft **630** may be hollow so as to define in blood pump **600** an internal lumen **670**.

(149) The second part, shown in FIG. 15B, is a housing or cannula **625** in which impeller **605** resides. Cannula **625** has a storage housing **660** for impeller **605** when the impeller is in a compressed state. Storage housing **660** may be nonexpandable. Alternatively, storage housing **660** may itself be expandable or cannula **625** may have an expandable portion for housing the impeller when the impeller is in its operational or deployed configuration. Whether there is a difference in the location of impeller **605** in its operating and stored configurations depends on whether the impeller is moved axially within cannula **625** for deployment or whether cannula **625** expands in the area in which the impeller is stored.

(150) The third part, shown in FIG. 15C, is a retainer sheath **700** which holds at least a portion of cannula **625** in a compressed state for insertion into a vessel of a patient. Each of these parts will be described more fully below.

(151) The cannula **625** of blood pump **600** has a nonexpandable portion **623** at its proximal end and an expandable portion **626** at its distal end. The expandable portion **626** may be flared at one or both ends to aide in fluid flow. The nonexpandable portion **623** of cannula **625** may be formed from conventional biocompatible polymer tubing and the like. The expandable portion **626** of cannula **625**, on the other hand, may be formed from a mesh **631**, such as a metal or polymer mesh, and an elastomer coating **633**. The mesh predominantly defines the radial stiffness and bending characteristics of the cannula, while the elastomer coats the mesh to form a continuous duct having a fluid-carrying capability.

(152) Mesh **631** may be in the form of a hexagonal cell matrix, or may include circumferential rings **692** and axial connectors **694**, as shown in FIG. 16. The circumferential rings predominantly control the radial characteristics while the axial connectors affect axial stiffness and bending performance.

(153) Mesh **631** may be formed from a flexible material, such as a polymer, metal, any shape memory material, or other material, and may include a machined metal cylinder with laser cut voids, a matrix of woven wires, or other configuration. Where mesh **631** is made from a memory metal alloy, such as nitinol, a constant diameter tube of the metal, having a metal thickness on the order of thousandths of an inch, for example, a thickness in the range of 0.005-0.007 inch, may be cut using a laser so as to leave a mesh structure. The constant-diameter mesh may then be expanded/contracted radially to the desired shape using a mandrel, and optionally a clamping mechanism may be used to ensure the mesh conforms to the mandrel geometry. The material is “shape set” to this configuration using, for example, heat treatment. The mandrel, and hence the diameter profile of the expandable portion **626** of cannula **625**, optionally can be customized to a particular patient. Alternatively, mesh **631** may be formed from a polymer. Other suitable materials for mesh **631** include other metals (such as alloys, including other memory metal alloys), polymers, other shape memory materials, and the like.

(154) Use of the laser-cutting and shape-setting steps enables complicated geometric patterns to be formed from the constant-diameter tube. An example cannula design may include a bell-mouth inlet (to minimize hydrodynamic losses), a hydrodynamic diffuser at the outlet (for pressure recovery from fluid velocity), a screen-like device at the inlet end (for avoidance of inlet flow obstructions), and additional material at the screen tip that serves as a dilator when the cannula is contracted.

(155) Once mesh **631** has been formed, a coating, such as elastomer coating **633**, may be applied to the mesh inner surface, outer surface and/or interstitially. The coating (which may be, for example,

biocompatible, corrosion resistant and/or flow improving) may be formed by a solution casting method or by other techniques known in the art, including forming the coating as a separate tube, fitting it over the mesh and heat shrinking it to produce a tight fit. An elastic polymer such as Elastane™ or Biospan™ may be used for coating **633**, as may other polyurethanes, or other polymers. Mesh **631** and coating **633** may provide a flexible, expandable portion **626** of cannula **625** that is a conduit for fluid flow. The expandable portion **626** of cannula **625** may be generally cylindrical with a flow inlet **642** at its distal end and a flow outlet **644** at its proximal end.

(156) The mesh **631** is radially expansible in a way which imparts a minimal length change (along the axial direction) during radial expansion/contraction. The expandable portion **626** of cannula **625** may radially contract or expand using stored potential energy, and thus is preferably a self-expanding/self-contracting device.

(157) The radial stiffness of the expandable portion **626** is controllable via the mesh thickness and the geometric density of the cell structure, which can vary along the cannula length. Such variability is useful to match the cannula stiffness with the imposed hydrodynamic loading, enabling a nearly constant radial deflection of the tube when operating as a flow duct (wherein the hydrodynamic pressure varies along the length). This is important in the region of the impeller to provide a constant operational tip gap.

(158) Cannula bending stiffness is also a controllable parameter that may vary axially. For example, where circumferential rings **692** and axial connectors **694** are used to form mesh **631**, the bending stiffness is predominantly controlled by the number and placement of the axial connectors, but also depends on the stiffness of the circumferential rings and the stiffness of the elastomer coating **633**. The relative placement of the circumferential rings largely affects the radial stability of the cannula during bending. For example, as shown in FIG. **16**, mesh **631** may have a substantial amount of interleaving of adjacent circumferential rings. This configuration yields a very stable cannula with respect to radial buckling caused by a bending deflection. Conversely, a mesh pattern with no interleaving yields a cannula that is prone to radial buckling during a bending deflection. Radial stiffness may be augmented via mesh thickness or mesh density. A dense mesh exhibits greater radial stability than a less dense mesh.

(159) FIG. **17** depicts cannula **625** in the expanded state. The expanded portion **626** of cannula **625** includes a distal end **646** having an inlet **642** through which blood enters the cannula, and a proximal end **648** having an outlet **644** through which blood leaves the cannula. The portion between inlet **642** and outlet **644** is the expandable portion **626** of cannula **625**. Inlet **642** may be provided with a plurality of inlet struts **652** which prevent obstructions from entering the cannula. Similarly, outlet **644** may be provided with a plurality of discharge struts **654** which act as stationary stator blades and remove swirl velocity from the discharge flow of impeller **605**. Inlet struts **652** and discharge struts **654** may occupy a short section of the cannula assembly (such as 1 cm) and may be flat linear elements arranged in a uniform circular disposition about the central axis of the device or may be part of mesh **631**. Alternatively, struts **654** may be formed with airfoil type cross-sections. Impeller **605** is located close to the proximal end **648**, and a guide wire **680** extends through cannula **625** and through hub **610** of impeller **605**. The blood flow through the expanded portion **626** of cannula **625** is from right to left (as shown in FIG. **17**) for an LVAD, blood entering the device through the distal end **646** and leaving the device through the proximal end **648**.

(160) An uncut region of the original tube retains the original tube diameter and may be used as the storage housing **660** in the form of a cup for retaining impeller **605** in the stored configuration. Storage housing **660** may be referred to as a non-expanded portion of cannula **625** through which blood does not flow. In this case, the expandable portion **626** of cannula **625** is attached to storage housing **660** through the discharge struts **654**. Alternatively, the cannula mesh **631**, discharge struts **654** and storage housing **660** may be formed from a single tube formed from the same. Hence, the cannula can and also be various attached components can be manufactured from a single piece of

tube, for example from a nitinol tube using laser cutting, with a mandrel used for shaping the mesh portion. Alternatively, various portions of the cannula can be manufactured separately and attached together using welding or other attachment techniques. Storage housing **660** may have a flared end **649** which may be defined by the shape of discharge struts **654** to aid in moving impeller **605** back to its stored position.

(161) Impeller **605** may be held in storage housing **660** in the stored configuration and moved axially into expandable portion **626** for deployment, such as by using drive shaft **630** to urge the impeller out of the storage housing. Impeller **605** then unfolds into the deployed configuration using the stored potential energy of blades **612** in the stored configuration. Alternatively, impeller **605** may be held in the stored configuration in the expandable portion **626** while contracted, and may deploy automatically upon expansion of the expandable portion. In still other embodiments, storage housing **660** itself may be expandable, allowing impeller **605** to expand to its deployed diameter without axial movement.

(162) In an example of cannula **625** described above, the expandable portion **626** thereof was formed from a nitinol tube having an inner diameter of 2.62 mm, an outer diameter of 3.02 mm, and a length of 150 mm. In the expanded state, portion **626** had a nominal inner diameter of 6.6 mm in the expanded section, and a nominal length of 133 mm. The expandable portion included 35 circumferential rings **692**, four axial connectors **694** per ring in a fully connected region (involving eight circumferential rings), and one axial connector **694** per ring in a minimally connected circumferential rings). Each region (involving twenty-eight circumferential ring **692** had four waves per ring, with a wave amplitude of 5.05 mm (at cut diameter). The interleaved fraction of the rings was $2.05/5$ =(where the interleaved fraction for fully interleaved is 1, the interleaved parameter being the overlapped distance divided by the wave amplitude). Finally, the typical thickness of inlet struts **652** and discharge struts **654** was 0.2 mm.

(163) A rotatable drive shaft **630** provides rotational coupling between a motor (not shown), located outside of the patient, and the impeller **605**. Drive shaft **630** may have a substantially rigid portion **632** at its distal end which is connected to impeller **605**, and a substantially flexible portion **634**. The flexible portion **634** of the drive shaft may be housed within a flexible tube **638** which supports the flexible portion and maintains its shape as it is driven rotationally. The proximal end of drive shaft **630** may be connected to the motor for rotating the drive shaft and with it impeller **605**. Alternatively, drive shaft **630** may be omitted, and the electric power may be provided through a proximal portion of the assembly to operate a pump motor and impeller **605**.

(164) Drive shaft **630** may have a diameter on the order of 1.1 to 2 mm, and may be hollow to allow guide wire **680** to pass therethrough. The flexible portion **634** of drive shaft **630** may be formed from a metal or polymer braid which is easily bendable so as to achieve a bend radius on the order of 1 cm. Commercially available flexible impeller drive shafts may be used in blood pump **600**, such as those formed from metal wire construction. However, heating problems due to friction between rotating and non-rotating components may occur within any small radius bends required for operation. A composite flexible shaft may be used to reduce such heating problems. The heating problem also can be addressed by providing lubricating or low-friction films on one or both of adjacent surfaces with high relative rotational motion. In that regard, a preferred drive shaft **630** may be constructed from coiled stainless steel, with an optional polymer support tube. A particularly preferred polymer is polytetrafluorethylene.

(165) The rigid portion **632** of drive shaft **630** may be supported by one or more bearings **672** retained in a bearing housing **675**. A saline solution may be directed into bearing housing **675** through internal lumen **670**, and the bearing unit end seal **674** may be dimensioned so that a of clean saline solution is infused (approximately 1-2 cc/hr). This fluid very small quantity into the patient flow helps clean impeller **605** and dampens drive shaft vibrations. The fluid flow may also prevent blood from entering bearing housing **675** and compromising its operation and life. If the density of drive shaft **630** is approximately the same as that of the saline solution or other

introduced fluid, most of the vibration can be damped. Drive shaft **630** may be formed from carbon or other fiber and polymer composite which has a lower density than metal and more closely matches the density of the saline solution. Other lower density draft shafts and/or higher density fluids may be used for vibration damping. The saline solution or other fluid may be introduced to bearing housing **675** through openings **678** in hollow drive shaft **630**.

(166) FIGS. **23A** and **23B** show an embodiment of blood pump **600** having an axially slidable storage housing **660**. As can be seen in these figures, bearing housing **675** may have a reduced diameter portion **677** between its ends housing bearings **672**. This reduced diameter portion thus defines a longitudinal space for sliding movement of an internal rib **662** defined by an indented annular channel **664** in storage housing **660**. In the deployed condition shown in FIG. **23A**, storage housing **660** has been moved proximally by the maximum extent permitted by the engagement of internal rib **662** with a proximal shoulder of bearing housing **675**, thereby revealing blades **612** of impeller **605** for deployment. In the stored configuration shown in FIG. **23B**, on the other hand, storage housing **660** has been moved distally to the maximum extent permitted by internal rib **662** contacting a distal shoulder of bearing housing **675**. In this position, the distal end of storage housing **660** surrounds blades **612** of impeller **605**, retaining them in the stored configuration.

(167) The internal lumen **670** of blood pump **600** receives guide wire **680**. Together, lumen **670** and guide wire **680** assist in positioning blood pump **600** within the patient. Guide wire **680** may have a two-part structure to assist in threading the guide wire through the bearing/seal assembly and through the hub **610** of impeller **605**, since part of that threading may be accomplished under factory controlled conditions, rather than at the time of use. In one embodiment, guide wire **680** may have a J-tip **682** which facilitates navigation of the tortuous arterial pathway from the femoral insertion site to the cardiac left ventricle chamber. Guide wire **680** may include an optional device on its proximal end to allow the attachment of a similar diameter extension of the guide wire, residing in the collapsed cannula **625**. The lumen **670** in blood pump **600** may have a relatively large diameter relative to the diameter of guide wire **680**. Guide wire **680** may have one or more additional distal end features such as a spherical shape, or a valve plug **689** to plug a hole in impeller **605** after withdrawal of the guide wire (see FIG. **22A**).

(168) Optionally, the guide wire channel extending through impeller **605** and the bearing unit end seal **674** may have a valve action, as is known in the art, sealing the guide wire passage after removal of guide wire **680**. Guide wire **680** may leave a mechanical seal, not shown, upon removal, or the material of impeller **605** may be designed so as to close the opening into lumen **670** upon removal of the guide wire. This avoids excess saline infusion into the patient.

(169) Blood pump **600** may be inserted into the patient's body using a sheathless insertion procedure. Such procedure may employ a retainer sheath **700** having a distal portion **702** and a proximal portion **704**, as shown in FIG. **15C**. Distal portion **702** may be about 20 cm in length, and have an inner diameter of about 9 fr (3.0 mm) and an outer diameter of about 10.5 fr (3.5 mm). The inner diameter of the distal portion allows storage of the collapsed cannula/impeller assembly. The proximal portion **704** of retainer sheath may be about 1 meter in length with an outer diameter of about 9 fr. This proximal portion **704** may serve as the housing for the flexible portion **634** of drive shaft **630** and for the non-expandable portion **623** of cannula **625**.

(170) A "pre-parked" integrated insertion sheath **800** may slide over the proximal portion **704** of retainer sheath **700**. The outer diameter of insertion sheath **800** is preferably about the same as the outer diameter of the distal portion **702** of retainer sheath **700**, in this example about 10.5 fr. When the proximal end of the distal portion **702** of retainer sheath **700** is pulled up against the distal end **802** of insertion sheath **800**, a smooth transition is evident, as shown in FIG. **25**. Thus, the combined retainer sheath **700** and insertion sheath **800** may be inserted into the patient's femoral artery as a single entity. After the insertion sheath is fully inserted into the femoral artery, the proximal portion **704** of retainer sheath **700** may be pushed into the patient, pushing the distal end of the retainer sheath and its contents into the patient's left ventricle.

(171) The most distal end **710** of retainer sheath **700** may have a series of slots (not shown) that allow slight expansion of the retainer sheath distal end during removal of cannula **625**. As the expandable portion **626** of cannula **625** must be collapsed during this process, the funnel shape created by these slots and the subsequent bending of the material of the sheath facilitates the collapse of the expandable portion of the cannula. Alternate means may be provided to facilitate the recollapse of the expandable portion **626** of cannula **625**.

(172) The drive motor rotates drive shaft **630** without rotating cannula **625** or retainer sheath **700**. The operation of blood pump **600** is controlled and monitored by a control unit (not shown) which displays status and controls various functions. Sensors, such as a pressure sensor and a flow rate sensor, may be affixed to various regions of the patient and/or blood pump **600**.

(173) The control unit preferably displays rpm of the drive motor, patient blood pressures, blood flow rate, information as to the location of the blood pump in the left ventricle, saline infusion and discharge rates, saline infusion temperature, etc. A filter may also be provided to show the presence of debris or blood in the saline discharge stream. The heart rate and blood flow rate are useful to be able to reduce patient dependency on the machine during recovery.

(174) Detailed Description of the Deployment of the Heart Assist Device

(175) Blood pump **600** may be percutaneously inserted through the femoral artery and threaded toward the heart for use, for example, as a left ventricular assist device. Blood pump **600** may be inserted into a patient using conventional cannula insertion methods. The impeller **605** of the device is then expandable in situ to enable an increased blood pumping capacity compared to conventional non-expandable devices. This can eliminate the requirement for surgical intervention.

(176) Insertion may be accomplished using the Seldinger technique which is well known in the art and used daily by surgeons and interventional cardiologists. In such technique, an introducer needle (not shown) is inserted into the femoral artery and used to introduce guide wire **680**. Once guide wire **680** is in place, the needle is withdrawn. An optional predilator (not shown) can be used over guide wire **680** to open up the arteriotomy (an opening in the femoral artery) to a size appropriate for insertion of blood pump **600**.

(177) A guide wire extension **685** (see FIG. 26) contained within the collapsed cannula **625** may be attached to the attachment device at the proximal end of guide wire **680**. By fixing the guide wire extension at the proximal end of guide wire **680**, the entire assembly is moved along the guide wire through the femoral artery opening previously created. During this process, the attachment of guide wire **680** to the guide wire extension **685** enters the body of the collapsed cannula assembly. To facilitate insertion without an introducer sheath, the distal end of the device may be provided with a tapered distal end dilator **688**, shown in FIGS. 20 and 22. This may be a compressed form of the inlet **642** of cannula **626** in which the inlet struts **652** fold down into a tapered, closed configuration similar in profile to a conventional dilator tip.

(178) The blood pump **600**, in its collapsed state, is then threaded over guide wire **680** and inserted into the artery. Once blood pump **600** is positioned, guide wire **680** may be removed. A nose bearing or seal at the distal end of impeller **605** can then seal the guide wire opening through the hub **610** of the impeller. This allows saline solution to be injected into impeller **605** for cooling and lubrication purposes, as well as to prevent blood from entering the lumen **670** of blood pump **600**.

(179) At the time of insertion, the expandable portion **626** of cannula **625** and impeller **605** are collapsed and may be contained within retainer sheath **700**. As described above, the proximal end **704** of retainer sheath **700** may include an optional second integrated insertion sheath **800** which replaces the function of the separate introducer sheath when the device is positioned in place. If there is no "introducer" sheath present during the insertion of the assist device into the femoral artery, the process is referred to as sheathless insertion. In FIG. 25, a smooth transition from the diameter of insertion sheath **800** to the maximum diameter of retainer sheath **700** is shown.

(180) During the insertion of blood pump **600**, the transition region has effectively zero length and zero change in outer diameter. The transition region is the region where the enlarged distal portion

702 of retainer sheath **700** meets the end **802** of insertion sheath **800**. After the collapsed cannula assembly is inserted into the femoral artery, and the position of insertion sheath **800** is fixed at the patient boundary, the transition region will be located several centimeters past the femoral artery opening. At this point, the medical practitioner holds the exterior of insertion sheath **800** stationary and continues to push the retainer sheath assembly along guide wire **680**, until the distal end of the collapsed cannula and retainer sheath reside at the distal end of the guide wire, within the left ventricle cavity **900**.

(181) A possible location of the device for LVAD use is shown in FIG. **26**. The non-expandable portion **623** of cannula **625** extends from the descending aorta **905** and into the femoral artery, from which it exits the host body. Guide wire **680** may be advanced into the left ventricle **900** using an optional guiding catheter (not shown), and a fluoroscope may be used to establish proper positioning of the assist device.

(182) Once blood pump **600** is properly positioned, retainer sheath **700** may be retracted, by a dimension of about 15 cm for some embodiments, allowing the expandable portion **626** of cannula **625** to expand to the deployed configuration.

(183) The final step in deploying blood pump **600** involves pushing impeller **605** from its stored position within storage housing **660** and positioning it at a specified location within the expanded portion **626** of cannula **625**. This may be accomplished by applying a small force to drive shaft **630** while holding retainer sheath **700** at a fixed location. Retainer sheath **700**, in turn, holds the expanded portion **626** of cannula **625** at a previously fixed location. Once no longer restrained by storage housing **660**, impeller **605** expands to the deployed configuration due to the action of stored strain energy.

(184) The expandable portion **626** of cannula **625** may attain its deployed configuration through the action of stored strain energy (potential energy). This process reveals the cannula inlet **642**, the expanded portion **626** of the cannula and the cannula outlet **644**. Successful operation requires that the cannula inlet **642** reside in the left ventricle **900** of the heart and that the outlet **644** reside in the aorta. A fluid seal must exist where the cannula is proximate to the aortic valve, and the surface smoothness of the cannula is preferably such that clinically significant abrasion of the aortic valve is prevented. Also, distal end struts **652** form an inlet grid which prevents the inlet from becoming blocked by soft tissues within the left ventricle.

(185) Impeller **605** may be moved toward the distal end of cannula **625** which curves around through a valve into the left ventricle **900** of the heart, while the flexible portion **634** of drive shaft **630**, coupled to impeller **605**, extends outside of the body of the patient and is rotated by a drive motor. The non-expanded portion **623** of cannula **625** similarly extends through the femoral artery and outside of the patient. The inlet **642** and struts **652** at the distal end of cannula portion **626** allow substantially unrestricted flow of blood into the device, where it is driven by impeller **605** outside of the device through a discharge mesh or struts **654** at the proximal end **648** of cannula portion **626**.

(186) Other methods of expanding impeller **605** may be used. One possible alternative approach may include infusing a liquid or gas through a shaft to inflate impeller **605**. Another approach may use rotational forces to induce blades **612** to form a desired shape. The potential energy in the blades, particularly in the blade roots, may be used to deploy the blades into their unstrained position, and hydrodynamic forces may cause the blades to further deform into their operating configuration.

(187) The retainer sheath **700**, previously retracted, serves to fix the position of the entire assembly within the patient. Blood pump **600** is now deployed and ready for connection to supporting equipment and use.

(188) When the patient recovers and can be weaned from the necessity of using blood pump intervention, impeller **605** may be pulled back into an inactive compressed configuration in storage housing **660** or in a non-expandable portion **623** of cannula **625**, and the expandable portion **626** of

the cannula may be pulled into retainer sheath **700**. Slots (not shown) or an outward flare **649** (see FIG. **9**) may be provided at the distal end of storage housing **660** to assist in retraction of cannula **625**. Retainer sheath **700** may then be pulled into proximity with insertion sheath **800**, and the insertion sheath, retainer sheath and cannula within it may be removed from the patient through the original femoral artery site. Subsequently, the wound in the patient may be closed in a conventional fashion. The flow rate and pressure rise of blood pumped by blood pump **600** is greater than current non-collapsible devices of the same diameter, and the rate of blood damage (hemolysis) is maintained at a clinically acceptable low level. The use of the device as an RVAD is similar to that described above.

(189) The expansion feature of blood pump **600** is an advantage over non-expandable prior art devices. If the device were non-expandable, the maximum cross-section would be limited to approximately 3 mm to allow for percutaneous insertion. However, this cross-section is insufficient to achieve sufficient blood flow to maintain the health of the patient.

(190) Other applications of the device according to the present invention include providing additional blood flow to other organs, assisting the heart during operations, and the like.

(191) The expandable portion **626** of cannula **625** may be expanded by any desired method. In one approach, mesh **631** may expand in a radial direction when the expandable portion **626** is contracted along an axial direction, for example using a mesh **631** having a hexagonal structure. In this approach, by applying tension to guide wire **680** through bull nose grommet **684**, shown in FIG. **21A**, the expandable portion **626** of cannula **625** can be shortened in the axial direction, providing radial expansion into the expanded state. Where a shape-memory material is used for the expandable portion of the cannula, the cannula will achieve its expanded state as the shape memory material reaches a predetermined temperature, such as when the cannula is inserted into a patient's blood vessel. Both impeller **605** and cannula **625** in their stored configurations may be held within retainer sheath **700**, whereby both may be mechanically deployed or may self-deploy when removed from the retainer sheath.

(192) Cannula **625** may have at least two configurations, including a stored configuration and an expanded (deployed) configuration. When used as part of a blood pump, cannula **625** in the deployed configuration may be about 20-30 cm long with a diameter of about 6-7 mm. In the stored configuration, cannula **625** may have a diameter of about 3 mm, allowing nonsurgical insertion of blood pump **600** into a human subject through a femoral artery. The larger deployed diameter allows for higher fluid flow rates after insertion, and reduced friction pressure losses compared with a non-surgically inserted blood pump having a non-expandable cannula.

(193) An improved process for blood pumping within a living subject includes providing an expandable impeller, inserting the impeller into a patient in a stored configuration (for example, with a diameter of between about 3 mm and about 4 mm), positioning the impeller at a desired location within a blood vessel of the patient, deploying the impeller (for example, to a diameter of between about 6 mm and about 7 mm), and operating the impeller in an operating configuration at a Reynolds number of between about 1,000 and about 30,000, and preferably between about 2,000 and about 10,000. Higher Reynolds number operation and more efficient pump operation may be possible with higher rotation speeds, but may increase the destruction of structures within the pumped fluid, such as blood cells. The operating diameter of the impeller may be at least about 50% greater than the stored diameter. In other examples, the operating diameter may be at least about 100% greater than the stored diameter. For animals, components may be scaled according to the size of the animal.

(194) Novel configurations and material choice allow the improved device to be compressed for cannula insertion into a patient. Representative devices include an expandable impeller, and a cannula that is at least in part expandable, inside of which the impeller rotates. Both the expandable impeller and the cannula have stored states that allow cannula insertion into a vein or artery using non-surgical methods. After insertion and location of the device, the expandable impeller and

cannula expand into deployed states. The impeller can be driven through a flexible drive shaft from a drive motor external to the host, or using a motor proximate to the impeller, possibly integrated with a bearing system.

(195) Other Pump Applications

(196) Applications of the improved fluid pump designs described herein are not limited to ventricular assist devices. The improved cannula and impeller designs are useful for any application where a stored configuration having reduced diameter is useful for locating the pump at a desired location. For example, a fluid pump operating underground may be introduced into a pipe, channel, or cavity through an opening of lesser diameter, and operate at a diameter greater than that of the opening used. Applications of an impeller deploying within an expandable cannula include a collapsible fire hose with an integral booster pump, a collapsible propeller, a biomedical pump for a biological fluid, and the like.

(197) In other examples, impellers may also be formed from metal sheets, plastic and non-resilient materials, for example in foldable configurations. Deployment may include the use of motors or other mechanical devices to unfold blades, automatic deployment induced by centrifugal forces, and the like.

(198) Although the invention herein has been described with reference understood that to particular embodiments, these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A catheter blood pump comprising: a cannula comprising: a generally cylindrical, flexible, and expandable central portion having a proximal end, a distal end and a blood flow conduit extending between the proximal end and the distal end; a flexible inlet portion at the distal end through which blood enters the blood flow conduit when the central portion is expanded, the flexible inlet portion provided with a plurality of spaced apart inlet struts configured to prevent an obstruction from entering the blood flow conduit; and a flexible outlet portion at the proximal end through which blood leaves the blood flow conduit when the central portion is expanded, the flexible outlet portion provided with a plurality of spaced apart outlet struts operative to reduce a swirl velocity of blood leaving the blood flow conduit; wherein the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts are fabricated from a collapsible shape memory material.
2. The catheter blood pump of claim 1, wherein the cylindrical, flexible, and expandable central portion defines a longitudinal axis between the proximal end and the distal end, and the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts respectively extend obliquely to the longitudinal axis.
3. The catheter blood pump of claim 2, wherein the cylindrical, flexible, and expandable central portion has a first longitudinal length, wherein the inlet portion has a second longitudinal length, wherein the outlet portion has a third longitudinal length, and where the second and third longitudinal length are each substantially smaller than the first axial length.
4. The catheter blood pump of claim 2, wherein the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts extend as flat linear elements arranged in a uniform circular disposition about the longitudinal axis.
5. The catheter blood pump of claim 1, wherein the generally cylindrical, flexible, and expandable central portion includes a nitinol tube, and wherein the plurality of spaced apart inlet struts are integrally formed with and extend from the nitinol tube.
6. The catheter blood pump of claim 1, wherein the generally cylindrical, flexible, and expandable central portion includes a nitinol tube, the plurality of spaced apart outlet struts are integrally

formed with and extend from the nitinol tube.

7. The catheter blood pump of claim 1, wherein the flexible, and expandable central portion is radially expansible from a reduced diameter to an expanded diameter defining the blood flow conduit.

8. The catheter blood pump of claim 7, wherein each of the reduced diameter and the expanded diameter extend uniformly between the proximal end and the distal end.

9. The catheter blood pump of claim 1, wherein the cannula further comprises a generally cylindrical, non-expandable portion attached to the generally cylindrical, flexible, and expandable central portion.

10. The catheter blood pump of claim 9, wherein no blood flows through the generally cylindrical, non-expandable portion in an operation of the pump.

11. The catheter blood pump of claim 9, wherein the outlet portion extends adjacent the generally cylindrical, non-expandable portion when the generally cylindrical, flexible, and expandable central portion is expanded.

12. The catheter blood pump of claim 1, further comprising an impeller disposed in the generally cylindrical, flexible, and expandable central portion of the cannula, the impeller operable to pump blood through the blood flow conduit between the inlet portion and the outlet portion.

13. The catheter blood pump of claim 12, wherein a combination of the impeller and cannula are sized to be inserted into a heart of a patient.

14. The catheter blood pump of claim 1, further comprising a retainer sheath operable to cause at least the plurality of spaced apart discharge struts to collapse.

15. A catheter blood pump comprising: a cannula comprising: a generally cylindrical, flexible, and expandable central portion having a first end, a second end opposing the first end, and a blood flow conduit extending between the first end and the second end: a flexible inlet portion at the first end through which blood enters the blood flow conduit when the central portion is expanded, the flexible inlet portion provided with a plurality of separated inlet struts configured to prevent an obstruction from entering the blood flow conduit; and a flexible outlet portion at the second end through which blood leaves the blood flow conduit when the central portion is expanded, the flexible outlet portion provided with a plurality of separated outlet struts operative to reduce a swirl velocity of blood leaving the blood flow conduit; wherein the plurality of inlet struts and the plurality of outlet struts are fabricated from a collapsible shape memory material; and an impeller disposed in the generally cylindrical, flexible, and expandable central portion of the cannula, the impeller operable to pump blood through the blood flow conduit between the inlet portion and the outlet portion.

16. The catheter blood pump of claim 15, wherein the generally cylindrical, flexible, and expandable central portion of the cannula includes a nitinol tube, and the plurality of inlet struts and the plurality of outlet struts are integrally formed with the nitinol tube.

17. The catheter blood pump of claim 15, wherein the cylindrical, flexible, and expandable central portion defines a longitudinal axis between the proximal end and the distal end, and the plurality of separated inlet struts and the plurality of separated outlet struts respectively extend obliquely to the longitudinal axis.

18. The catheter blood pump of claim 17, wherein the cylindrical, flexible, and expandable central portion has a first longitudinal length, wherein the inlet portion has a second longitudinal length, wherein the outlet portion has a third longitudinal length, and where the second and third longitudinal lengths are substantially smaller than the first longitudinal length.

19. The catheter blood pump of claim 17, wherein the plurality of inlet struts and the plurality of outlet struts extend as flat linear elements arranged in a uniform circular disposition about the longitudinal axis.
